

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2023-24

Botany
Paper: BOB-101 — Cryptogams

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions, selecting at least two each from Sections A and B and Section-C which is compulsory. The figures in the right-hand margin indicate marks.

Section-A

Question 1. Discuss along with suitable diagrams the comparative features of the sporophytes of *Anthoceros* and *Funaria*. [15]

Question 2. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Gametophyte of *Marchantia*
- (b) False branching in cyanobacteria
- (c) Type of zoospores in *Draparnaldiopsis*

Question 3. Discuss the complete life cycle of *Ectocarpus*, along with suitable diagrams. [15]

Question 4. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Elaters and pseudoelaters in Bryophytes
- (b) Capsule of *Funaria*
- (c) Cystocarp

Section-B

Question 5. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Various types of Steles in Pteridophytes
- (b) Outline of Ainsworth's classification
- (c) General characters and types of fruiting body in Ascomycotina

Question 6. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Reproduction in *Selaginella*
- (b) Reproduction in *Pteris*
- (c) Parasexuality in fungi

Question 7. Describe the general characteristic features of Basidiomycotina and describe in brief the life cycle of *Puccinia*. [15]

Question 8. Discuss the life cycle of *Equisetum* along with suitable diagrams. [15]

Section-C

Question 9. Answer the following (not more than 50 words each): [10 × 1 = 10]

- (a) Why do brown algae appear brown in colour?
- (b) Who proposed the Stelar theory in Pteridophytes?
- (c) Write the name of a Coprophilous fungi.
- (d) Dolipore septum is a characteristic feature of which group of fungi?
- (e) Pseudoelaters are the characteristic feature of which bryophyte?
- (f) Name one algal genus that exhibits stephanokont condition.
- (g) Name the sterile tissue formed inside the capsule of *Anthoceros*.
- (h) Which structure helps in spore dispersal in *Equisetum*?
- (i) Which fungus is commonly known as bread mold?
- (j) Gonimoblast filaments are formed in which algal genus?

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